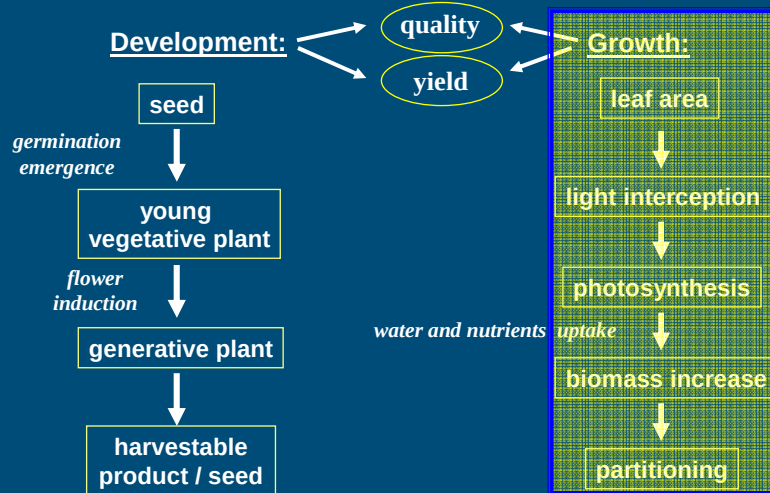


HPC 21306 Crop Ecology 2010

- course scheme -



Influencing factors: light (amount/quality/day length), temperature (level/day & night regime), CO₂, humidity, EC, water, nutrients, plant density, pruning.
Scaling up to higher integration levels - Interactions between climate factors and crop in a greenhouse.
Not considered: organic matter mineralisation, specific nutrients, pests & diseases, plant breeding.

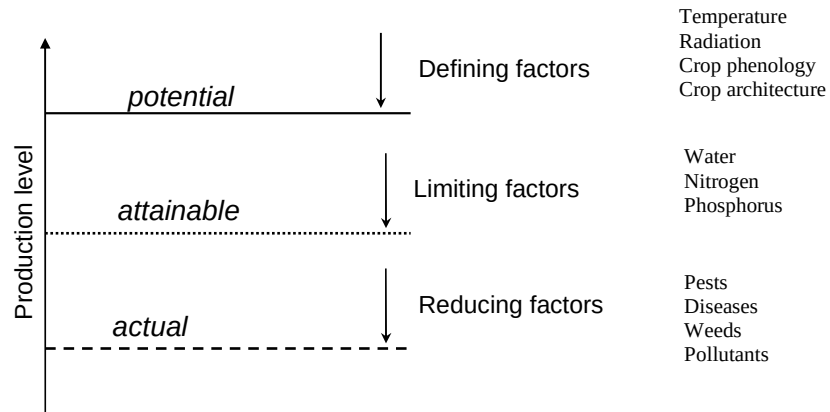
Lintul1: Potential crop growth

Chapter 10A, the math-model p.1-7 & the program p.7-12, p.30-32.
 Exercises on p.14

Peter Leffelaar
 Eelco Meuter

Crop Ecology (HPC-21306)

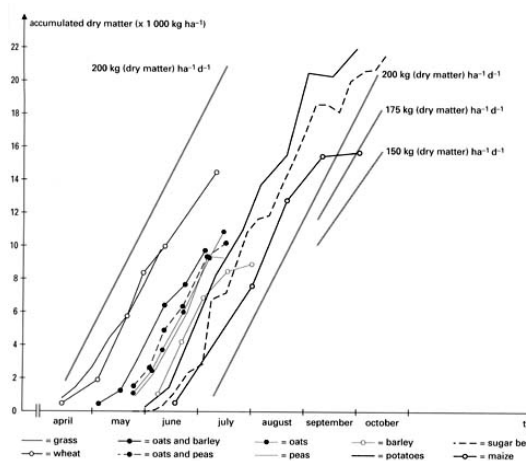
Potential, attainable and actual production



Good production situation
(good climate/ good soil)

From: Lövenstein, Lantinga,
Rabbinge and van Keulen, 1995

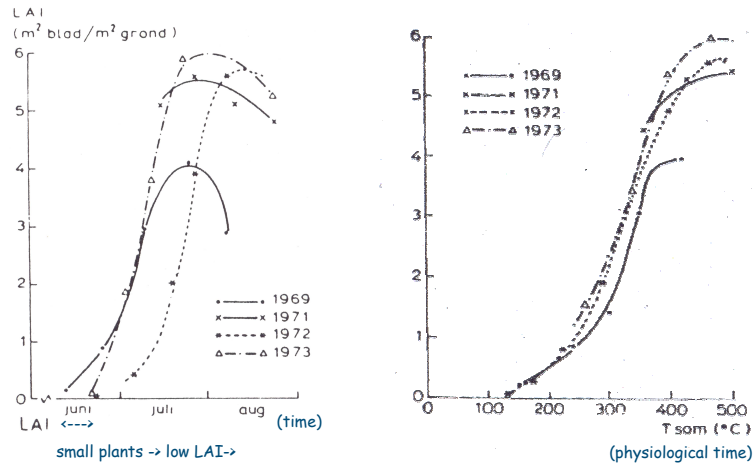
Optimal growth rates



Growth rates of the main agricultural crops in the Netherlands under
(near)-optimal growth conditions, compared to growth rates of 200, 175
and 150 kg (dry matter) ha⁻¹ d⁻¹
(Source: Sibma, 1968)

From: Sibma, 1968

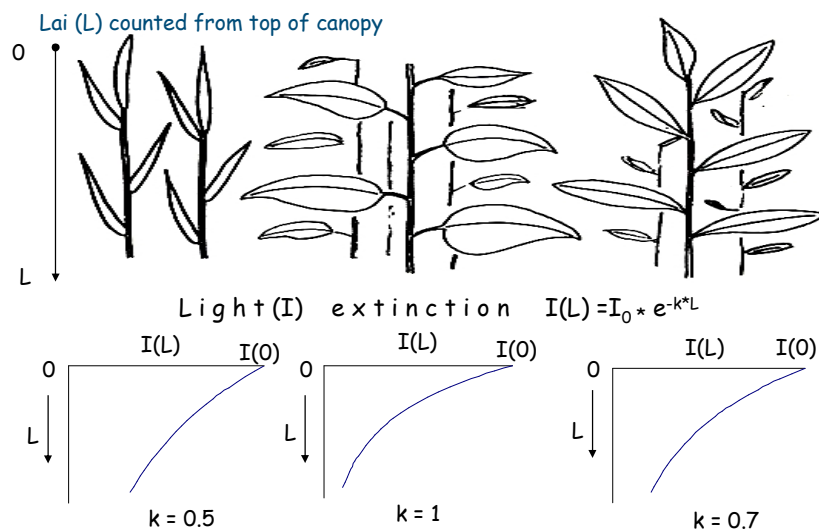
Leaf area of maize as a function of (physiological) time



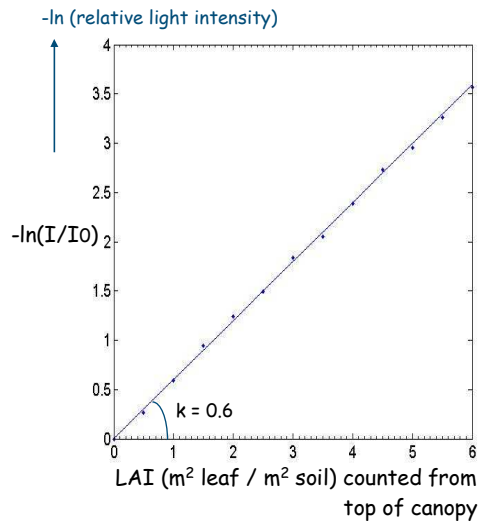
small plants → low LAI →
not all sunlight used →
growth rate << 200 kg/(ha*d)

From: Sibma, 1987

Light extinction is a function of LAI



Light extinction coefficient, k , follows from slope



$$I(L) = I_0 \cdot e^{-k \cdot L}$$

$$\Rightarrow -\ln\left(\frac{I}{I_0}\right) = k \cdot L$$

maize

From: Sibma, 1987

Spectral composition of global radiation

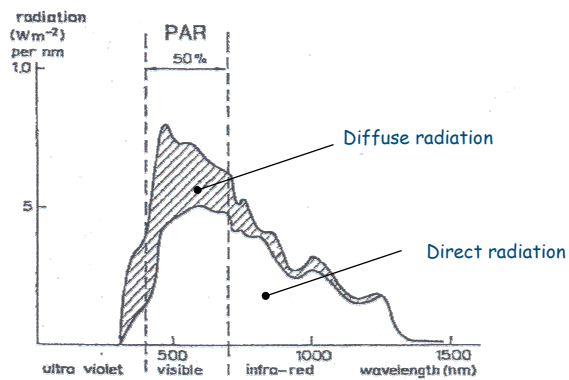
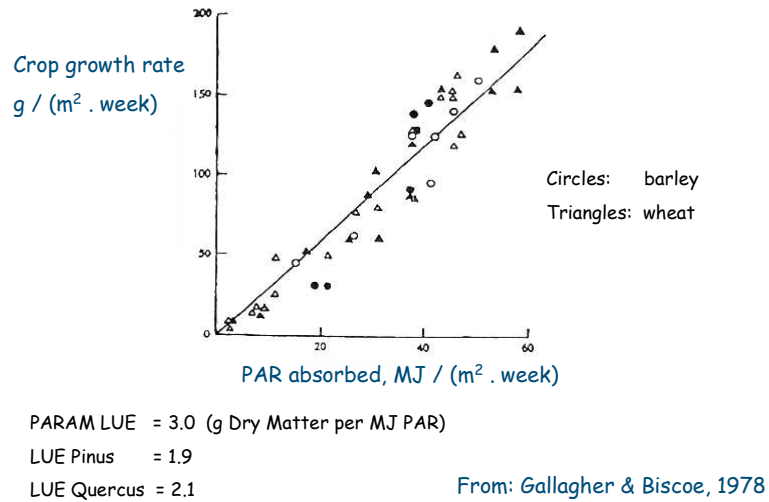


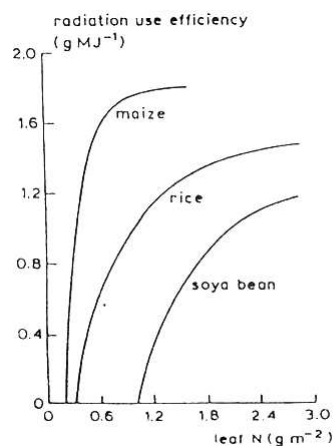
figure 2.2 Spectral composition of global radiation, discriminating between direct radiation (clear area) and diffuse (i.e. indirect) radiation scattered by clouds (hatched area). PAR = photosynthetically active radiation.
 (Source: Monteith and Unsworth, 1990, Principles of environmental physics. Edward Arnold, London, 291 p.)

From: Monteith & Unsworth, 1990

Light Use Efficiency, LUE, follows from slope



Radiation use efficiency as a function of leaf nitrogen content per unit area



From: Sinclair & Horie, 1989

Light interception, utilization and growth in LINTUL

TITLE LINTUL

INITIAL

INCON WTOTI = 5.

TIMER STTIME=1. ; FINTIM=100. ; DELT=1. ; PRDEL = 5.

TRANSLATION_GENERAL DRIVER = 'EUDRIV'

PRINT LAI, WTHA, PARINT

DYNAMIC

LAI = WTOTAL * LWR * SLA

PARINT = (0.5 * DTR) * (1. - EXP(-K * LAI))

$L = W F_{lv} S_{la}$

$I_{int} = 0.5 DTR (1 - e^{-kL})$

compare Eq.(13)

Eq.(2)

GTOTAL = LUE * PARINT

WTOTAL = INTGRL(WTOTI, GTOTAL)

WTHA = WTOTAL / 100.

$dW/dt = RUE I_{int}$

$W = \int dW/dt dt$

Eq.(3)

Eq.(4)

PARAM DTR = 20. ; LWR = 0.5 ; SLA = 0.022 ; LUE = 3.0 ; K = 0.6

END

STOP

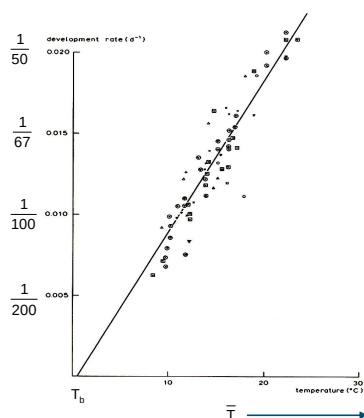
ENDJOB



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Development rate of spring wheat as a function of temperature from emergence till flowering . . .



$$\frac{1}{\text{\# days}} = \bar{T} \cdot 0.0009 \Rightarrow$$

$$\text{"temperature sum"} = \bar{T} \cdot \text{\# days} \Rightarrow$$

$$\frac{1}{0.0009} = 1110 \text{ } ^\circ\text{Cd}$$

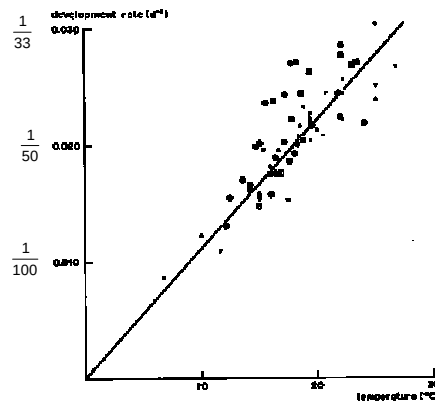
Data from over 20 papers assembled
by: Van Keulen & Seligman, 1987



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... and from flowering to ripeness



$$\frac{1}{\# \text{ days}} = \bar{T} \cdot 0.00103 \Rightarrow$$

$$\bar{T} \cdot \# \text{ days} = \frac{1}{0.00103} = 970 \text{ }^{\circ}\text{Cd}$$

Vegetative Phase: 1110 $^{\circ}\text{Cd}$

Generative Phase: 970 $^{\circ}\text{Cd}$

Whole growing season: 2080 $^{\circ}\text{Cd}$

finish condition Lintul

Data from over 20 papers assembled
by: Van Keulen & Seligman, 1987

Use of weather data in LINTUL

TITLE LINTUL

* 1. Initial conditions and run control
FINISH TSUM > 2080.

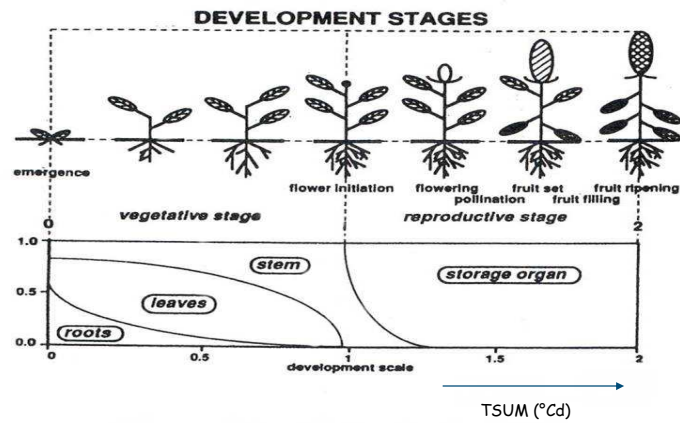
* 2. Environmental data and temperature sum
WEATHER WTRDIR='C:\SYS\WEATHER\'; CNTR='IT'; ISTN=3; ...
IYEAR=1985

DTR = RDD / 1.E+6
DAVTMP = 0.5 * (TMMN + TMMX)
DTEFF = MAX(0. , DAVTMP - TBASE)
EMERG = INSW(TIME-DOYEM , 0. , 1.)
TSUM = INTGRL(ZERO, RTSUM)
RTSUM = DTEFF * EMERG

$$T_{sum} = \int (T_a - T_b) dt \quad \text{Eq.(6)}$$

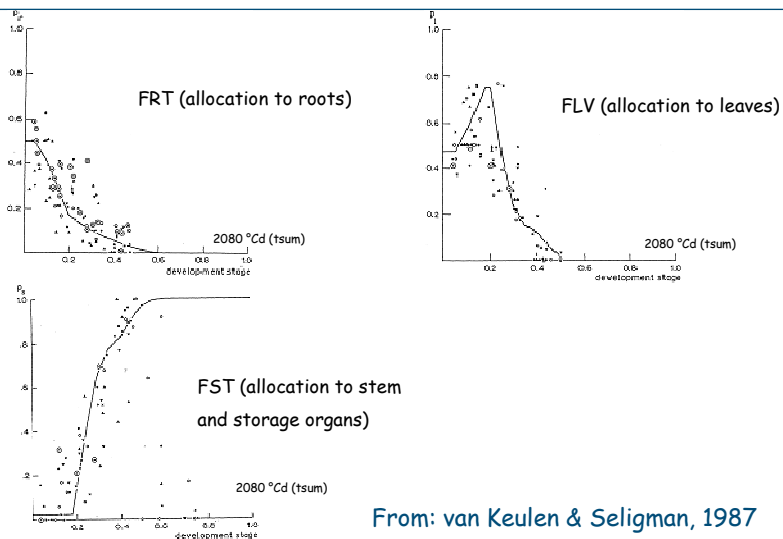
* 6. Functions and parameters for spring wheat
PARAM TBASE = 0.
PARAM DOYEM = 32.

Crop development stages



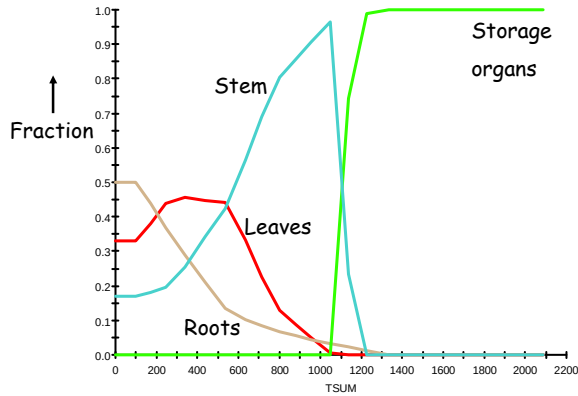
From: Lönstein, Lantinga, Rabbinge and van Keulen, 1995

Substrate allocation patterns in Spring Wheat: Measured



From: van Keulen & Seligman, 1987

Substrate allocation patterns in Spring Wheat: In model



Allocation pattern of assimilates for spring wheat
as used in Lintul (Light interception and utilization)



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Growth rates and dry matter production of plant organs

TITLE LINTUL

FRT = AFGEN(FRTTB , TSUM)
FLV = AFGEN(FLVTB , TSUM)
FST = AFGEN(FSTTB , TSUM)
FSO = AFGEN(FSOTB , TSUM)

WLVG = INTGRL(WLVI , RWLVG)
WLVD = INTGRL(ZERO , DLV)
WST = INTGRL(ZERO , RWST)
WSO = INTGRL(ZERO , RWSO)
WSO_{THA} = WSO / 100.
WRT = INTGRL(ZERO , RWRT)
WLV = WLVG + WLVD

RWLVG = GTOTAL * FLV - DLV
RWST = GTOTAL * FST
RWSO = GTOTAL * FSO
RWRT = GTOTAL * FRT

$$dW_{lv}/dt = RUE I_{int} F_{lv} - r_d W_{lv} \quad \text{Eq.(8)}$$

$$dW_{st}/dt = RUE I_{int} F_{st} \quad \text{Eq.(7)}$$

$$dW_{so}/dt = RUE I_{int} F_{so} \quad \text{Eq.(7)}$$

$$dW_{rt}/dt = RUE I_{int} F_{rt} \quad \text{Eq.(7)}$$



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Functions for spring wheat

TITLE LINTUL

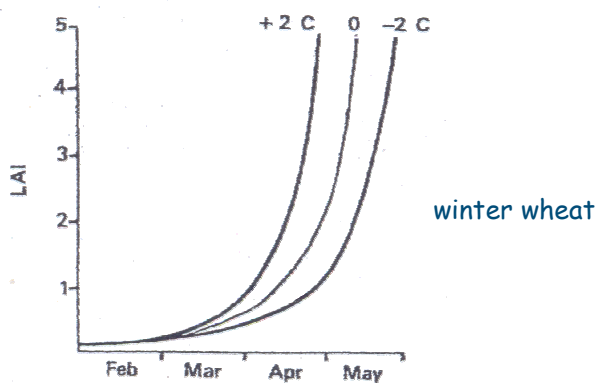
FUNCTION FRTTB = 0.,0.50 , 110.,0.50 , ...
 275.,0.34 , 555.,0.12 , 780.,0.07 , ...
 1055.,0.03 , 1160.,0.02 , 1305.,0. , 2500.,0.

FUNCTION FLVTB = 0.,0.33 , 110.,0.33 , ...
 275.,0.46 , 555.,0.44 , 780.,0.14 , ...
 1055.,0. , 2500.,0.

FUNCTION FSTTB = 0.,0.17 , 110.,0.17 , ...
 275.,0.20 , 555.,0.44 , 780.,0.79 , ...
 1055.,0.97 , 1160.,0. , 2500.,0.

FUNCTION FSOTB = 0.,0. , ...
 1055.,0. , 1160.,0.98 , 1305.,1. , 2500.,1.

Early leaf growth: temperature dependent



From: Monteith, 1966

Parameters for spring wheat & Growth of Leaf Area (GLA)

```
PARAM LAII = 0.012 ; SLA = 0.022 ; RGRL = 0.009 ; ...
      TSUMAN = 1110. ; LAICR = 4. ; RDRSHM = 0.03
FUNCTION RDRT = -10,0.03 , 10,0.03 , 15,0.04 , 30,0.09 , 50,0.09
```

```
*-----*
SUBROUTINE GLA(TIME,DOYEM,DTEFF,TSUM,LAII,RGRL,DELT,
$             SLA ,LAI,GLV,                      GLAI)
IMPLICIT REAL (A-Z)
```

```
*---- Growth during maturation stage:
      GLAI = SLA * GLV
```

$$dL/dt = RUE \cdot I_{int} \cdot F_{lv} \cdot S_{lo}$$

Eq.(7 & 13)

```
*---- Growth during juvenile stage:
      IF ( (TSUM.LT.330.) .AND. (LAI.LT.0.75) )
$      GLAI = LAI * (EXP(RGRL * DTEFF * DELT) - 1.) / DELT
```

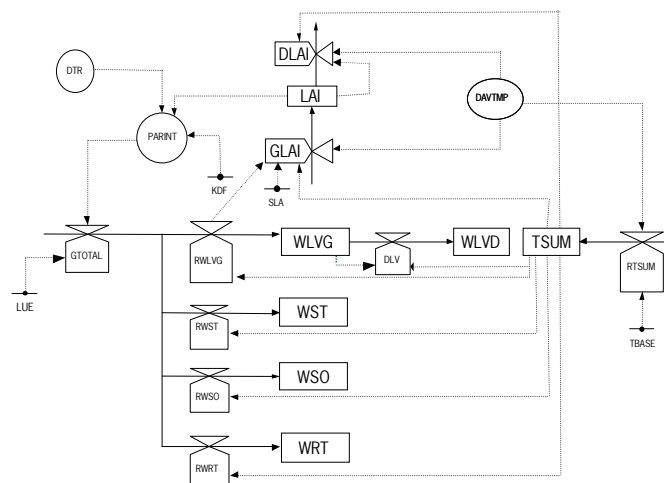
Eq.(12)

```
*---- Growth at day of seedling emergence:
      IF ( (TIME.GE.DOYEM) .AND. (LAI.EQ.0.) )
$      GLAI = LAII / DELT
```

```
*---- Growth before seedling emergence:
      IF (TIME.LT.DOYEM) GLAI = 0.
```

```
RETURN
END
```

Relational diagram of Lintul1

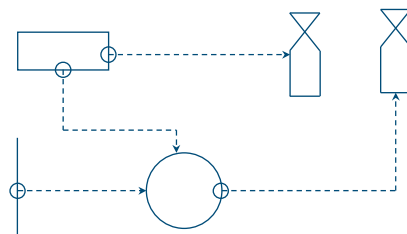


Forrester notation . . .

Symbol		Dimension
	Integral	Amount
	Differential equation (decision about rate)	Amount/time
	Flow of substance	Amount/time
	Discontinuous flow of substance	Amount/time
	Flow of information	No unit
	Input variable	Not known beforehand
	Auxiliary variable	Not known beforehand

. . . Forrester notation (continued)

Information flows can only be used:
 Information flows should point towards a
 place of decision, never to a state:



Practical Lintul1

Exercises on p. 10A14 on the following subjects:

- Consequence of weather and different emergence dates on crop growth
- The sensitivity of yield for some important crop parameters or, differently stated, the consequence of uncertainties in parameter values
- Effects of parameter changes in light extinction (light capture), and light use efficiency on growth
- Potential yield as a function of emergence dates

There are data provided for a dry, an average and a wet year

Choose between Brazil and the Netherlands

Prepare a report for yourself on the PC to prepare for the exam

References used in the slides

- Coombs, J., D.O. Hall, S.P. Long and J.M.O. Scurlock, 1985. Techniques in bioproductivity and photosynthesis. Pergamon International Library of science, Technology, engineering and social studies, Oxford, pp.298.
- Gallagher, J.N. & P.V. Biscoe, 1978. Radiation absorption, growth and yield of cereals. *Journal of Agricultural Science, Cambridge* 91: 47-60.
- Lövenstein, H., E.A. Lantinga, R. Rabbinge and H. van Keulen, 1995. Principles of Production Ecology, Syllabus of Chair Group Theoretical Production Ecology, pp.121.
- Monteith, J.L., 1966. Physical limitations to crop growth. In: *Agricultural progress*, Vol. 41: 9-23. Agricultural Education Association.
- Monteith, J.L. & M.H. Unsworth, 1990. Principles of environmental physics. Edward Arnold, London, pp.291.
- Sibma, L., 1968. Growth of closed green crop surfaces in the Netherlands. *Neth. J. of Agric. Sc.* 16: 211-216.
- Sibma, L., 1987. Ontwikkeling en groei van *Zea mays* L. onder Nederlandse omstandigheden. *Gewassenreeks 1*, Pudoc, Wageningen, the Netherlands, pp.57.
- Sinclair, T.R., and T. Horie, 1989. Leaf nitrogen, photosynthesis and crop radiation use efficiency: a Review. *Crop Science* 29: 90-98.
- Sinclair, T.R. and Muchow, R.C., 1999. Radiation use efficiency. *Adv. Agron.* 65, 215-265.
- Van Keulen, H. & N.G. Seligman, 1987. Simulation of water use, nitrogen nutrition and growth of a spring wheat crop. *Simulation Monographs*, Pudoc, Wageningen, pp. 310.